Skill Compass

Generative AI for the SDGs Hackathon · Group 12 · Gülşen Ebrar Cırık, Yiğit Emre Demir, Eray Irmak, Tamer Berksu Tekkaya, Berkay Mutlu

1. PROBLEM AND SDG LINK

Many young people with no formal work experience believe they have “nothing to put on a CV”. In practice they have often run the till in a family shop, tutored a sibling, led a team or volunteered. They do not know how to describe this in employers' language, which skills their target field needs, or what to learn first. Career guidance that could help them is unevenly available, and generic AI CV tools tend to make things worse by inventing achievements.

- **SDG 8.6:** substantially reduce the proportion of youth not in employment, education or training.
- **SDG 4.4:** increase the number of youth with skills relevant for employment.

Target users: vocational-school graduates, NEET youth and students without internships. Secondary users are school counsellors and youth-centre volunteers who help them.

2. SOLUTION

The user writes what they have done in everyday language, names a target field and can optionally paste a job ad. The tool returns:

- skills they already have, each pinned to a verbatim quote from their story;
- what the field needs, grouped into **technical skills, tools and software, soft skills, knowledge and qualifications, and experience**, each marked *have / partly / to build* with a concrete next step;
- a three-stage roadmap (next 2 weeks, 1–3 months, 3–6 months);
- CV lines and a STAR interview story built only from facts in the story, plus questions that would strengthen the profile.

Original contribution: an AI career tool that cannot inflate a CV. Every claim is checked in code, independently of the model. Skill quotes must be found in the story. A “you have it” status without a story quote is downgraded to *Unverified*. Every number in a CV line must appear in the story. With an ad, each requirement must quote the ad; without one, requirements are labelled “general guidance”. The copy button copies only the CV lines that passed the checks.

3. MODEL AND ARCHITECTURE

The interface is two short steps on one phone-first page, in English or Turkish. One LLM call returns structured JSON. The public version uses the free tier of Google's Gemini API, and the code also supports Claude; deterministic JavaScript then runs the checks and shows a short result with details folded away. A small server function on Vercel's free plan holds the API key, so **anyone can use the public link without an account or key, at zero cost**. Inputs are length-limited and rate-limited, and nothing is stored.

4. KEY PROMPTS AND DESIGN RATIONALE

Technique	Where and why
Chain-of-thought (`reasoning` field, first)	Separates “what the person did” from “what employers value” before any claim is written. The reasoning is visible in the UI.
One-shot example	A single example of the evidence format (an everyday sentence becomes a named skill with a verbatim quote). We used one example because more would bias the model toward the example's sector.
Zero-shot	For field requirements and the roadmap, where the model's general knowledge is the value.
Hard constraints	Verbatim, untranslated quotes; no number that is not in the story; at most 3 “general” requirements when an ad is given; hard filters such as a required degree stated honestly but with a route around them; resource types instead of specific websites, which are a common hallucination.

The full prompt and rationale are in `prompts/skill_mapping.md`.

5. CHALLENGES

- **Encouraging vs. honest.** The audience needs confidence, but false confidence costs them interviews. We kept the tone positive in the headline and the skills, and strict in the checks.
- **Requirements without an ad.** These cannot be verified against a source. Instead of hiding this, the UI labels them and tells the user to compare them with real ads.
- **Quotes across languages.** The story may be in Turkish and the answer in English, so quotes are kept in the original language for the checks.

6. ETHICAL ASSESSMENT

Risk	Who is affected	Mitigation
Invented skills or inflated CV numbers	The user, who may be caught out in an interview, and employers	Quote and number checks in code; only verified lines can be copied.
Outdated or wrong field requirements	Users without an ad	Labelled “general guidance”; users are told to compare with several ads; pasting an ad gives checked requirements.
Biased advice (gender, origin, disability, family background)	Disadvantaged youth	Explicit prompt rule. Next step: test the same story with only the name or gender changed and compare outputs.
Discouragement (“you do not meet the requirements”)	Low-confidence users	Every gap comes with a concrete, free or cheap next step and a route around hard filters.
Privacy of personal stories	All users	Our app stores nothing, but free-tier AI providers may use inputs to improve their products. The page tells users not to include names, ID numbers or addresses, and inputs are length-limited.

7. HOW WE KNOW IT WORKS

- **Accuracy:** on a set of 15–20 stories, measure the share of skills and CV lines that pass the checks, and have a reviewer rate whether the listed skills are reasonable.
- **Impact:** ask young users to write a CV with and without the tool, then have a recruiter or counsellor rank them blind. Also ask users before and after whether they know their next step.